GAIRA V7

The Runtime Platform

Phase 10 — frozen science, one runtime, five surfaces, identical answers

Status	COMPLETE — packaging phase, science unchanged
Engine	50 LSMs · 49 CSMs · 154 molecules · 16 chemistry axes
Atlas fingerprint	2e43ddcca7d3be41c5f9da016fb8277f
Frozen assets pinned	10 content digests, all verified
Cross-surface parity	60 comparisons across 6 surfaces · 0 divergent
Maximum discrepancy	0.0e+00 (tolerance 1e-12)
Phase 09 science	reproduced at max deviation 0.0e+00
molecule top-1 / top-5 / MRR	0.6053 / 0.7947 / 0.6870
Single-spectrum latency	2.34 ms median (p95 3.11 ms)
API / MCP overhead	1.61 ms / 2.97 ms
Engine load	0.279 s
Gates	17 of 17 PASS
Test suite	1436 passed · 1 skipped · 0 failed
Surfaces	Python SDK · CLI · FastAPI · MCP · Streamlit
Dependencies	no LLM · no cloud account · no network · no external volume
Scope	pure Raman reference spectra only

Chemistry Evidence is RELATIVE — not a concentration, not an abundance, not a mixture fraction. Retrieved molecules are reference analogues, not identifications.
Sources of record: PHASE_10_CONTEXT.md · PHASE_10_ENGINE_FREEZE_AUDIT.md · PHASE_10_ARCHITECTURE.md · PHASE_10_API_SPEC.md · PHASE_10_MCP_SPEC.md · PHASE_10_STREAMLIT_
PHASE_10_PLUGIN_ARCHITECTURE.md · PHASE_10_DEPLOYMENT_GUIDE.md · PHASE_10_VALIDATION_REPORT.md · PHASE_10_SCIENTIFIC_AUDIT.md · PHASE_10_DECISION_GATE.md

Contents

This document explains what GAIRA V7 is, what is frozen, how a raw Raman spectrum becomes an inference, where each consumer surface fits, why no language model is required, and how future modality and sample-context plugins would extend the platform without touching the science.

It is written to be read start to finish by someone who is not a spectroscopist. Technical sections are followed by a plain-English gloss marked in blue.

What GAIRA V7 is

A frozen biochemical coordinate system for Raman spectroscopy, and the runtime that makes it reachable without letting any route change an answer.

PART I

The problem, and what GAIRA answers

Shine a laser at a sample and a small fraction of the light returns with its colour shifted. The size of each shift is set by how the molecules present vibrate, so the pattern of shifts is a fingerprint of the chemistry. That pattern is a Raman spectrum.

Why the textbook approach fails

A textbook treats a spectrum as a fingerprint to be matched against a library. Real samples do not cooperate, and three facts govern this architecture:

— Spectra are mixtures, not fingerprints. What reaches the detector is the sum of everything in the illuminated volume.

— A peak is not a molecule. A band near 1450 cm-1 says a CH2 group is bending, and thousands of biological molecules contain CH2 groups.

— Nearby is not the same. Published assignments frequently match a peak to a molecule because the wavenumbers agree within a few units and the biology is plausible. That is not evidence.

What GAIRA answers instead

Rather than 'which molecule is this?', the engine answers a question it can support: what chemistry does the evidence favour, how strongly, and by way of which specific spectral features? It still returns ranked molecular candidates, because a shortlist is useful, but the load-bearing output is the chemistry reading.

> **In plain terms**
>
> *Think of a doctor reading a blood panel. They rarely conclude 'this is precisely molecule X at precisely concentration Y'. They read a pattern and say which processes it is consistent with, and how sure they are. GAIRA is built to do the second thing well rather than the first thing badly.*

PART I

What is frozen, and how

The scientific architecture was frozen at the end of Phase 09. Phase 10 may not change it, and the freeze is enforced by code rather than by convention.

What is frozen

canonical preprocessing	450–1800 cm-1 · 2.0 step · 676 bins · asLS → Savitzky-Golay(9,3) → L2
LSM dictionary	50 non-negative motifs, learned within chemistry class
CSM dictionary	49 consensus motifs — THE canonical representation
retrieval	cosine over a 154-molecule bank
Chemistry Evidence	16 axes, model D:A_max_idf with $\lambda = 0.5$
calibration	temperature scaling, $T = 0.4538$
confidence and warnings	Phase 05 thresholds, untouched

Two independent verification layers

Phase 09 verifies four DECLARED fingerprints — values recorded inside PHASE_STATE.json and csm_registry_v1.json by the phases that produced them. That check answers 'did the producing phase claim this artefact'. It does not answer 'is the file on disk still the file that phase wrote', because the declared fingerprint lives in a different file from the artefact.

Phase 10 adds the second layer: the content digest of every one of the ten files the engine opens, recomputed from the committed tree and verified before the runtime will serve anything. The loader was instrumented to find those ten — every one is inside the repository and git-tracked.

In plain terms

Together these answer two different questions. Change a dictionary in place and the declared fingerprint would not notice; change which artefact a phase claims and the content digests would not notice. Both layers run.

Why this matters

Consequence for deployment: an external volume is NOT required. The frozen atlas is about ten megabytes of committed files, so a clean clone can run inference immediately.

PART I

How a spectrum becomes an inference

One path, no branches, no parameter tunable at inference time.

1 · Parse

CSV, TSV, two-column text or arrays. Delimiter, header, column identity and axis direction are detected, and every decision is reported as a diagnostic rather than applied silently.

2 · Validate

Three severities. ERROR stops the run — too few points, no usable overlap with the canonical window, an all-zero or constant spectrum, an unsupported modality. WARNING runs with a stated limitation. INFO records scope.

3 · Preprocess

Crop, resample to 676 bins, remove the fluorescence background by asymmetric least squares, smooth, normalise to unit length. The last step discards absolute intensity, which is why no output can be read as a concentration.

4 · Project

Non-negative least squares onto 50 local motifs, then onto the 49 consensus motifs. Those 49 numbers are the canonical representation and the only thing any later stage reads.

5 · Retrieve

Cosine similarity against 154 reference molecules. Because a cosine is an inner product, each score decomposes exactly into per-motif contributions — which is what makes the ranking explainable rather than merely produced.

6 · Chemistry

Collapse the activation onto 16 chemistry axes with a hierarchical model that lets coarse chemistry gently inform fine chemistry without ever excluding it. Then calibrate.

7 · Report

Confidence, audit, provenance and a deterministic interpretation paragraph. No language model is involved.

> **In plain terms**
> *Each stage answers a narrower question than the last: can this run, is it clean, what patterns is it made of, which of those are real and distinct, what does it resemble, and what does that mean chemically.*

The platform

Twelve figures. Each is preceded by a page explaining what it shows and what it does not license you to conclude.

FIGURE 1

The shape of the platform

Phase 10 adds no science. It takes the engine frozen at the end of Phase 09 and makes it reachable — from Python, from a terminal, over HTTP, through MCP, and from a browser — without letting any of those routes change an answer.

The blue band at the top is the frozen engine. It preprocesses a spectrum, projects it onto 50 local motifs and then 49 consensus motifs, retrieves the nearest of 154 reference molecules, aggregates 16 chemistry axes, calibrates a confidence and builds a provenance chain. It is the only thing in this diagram that computes.

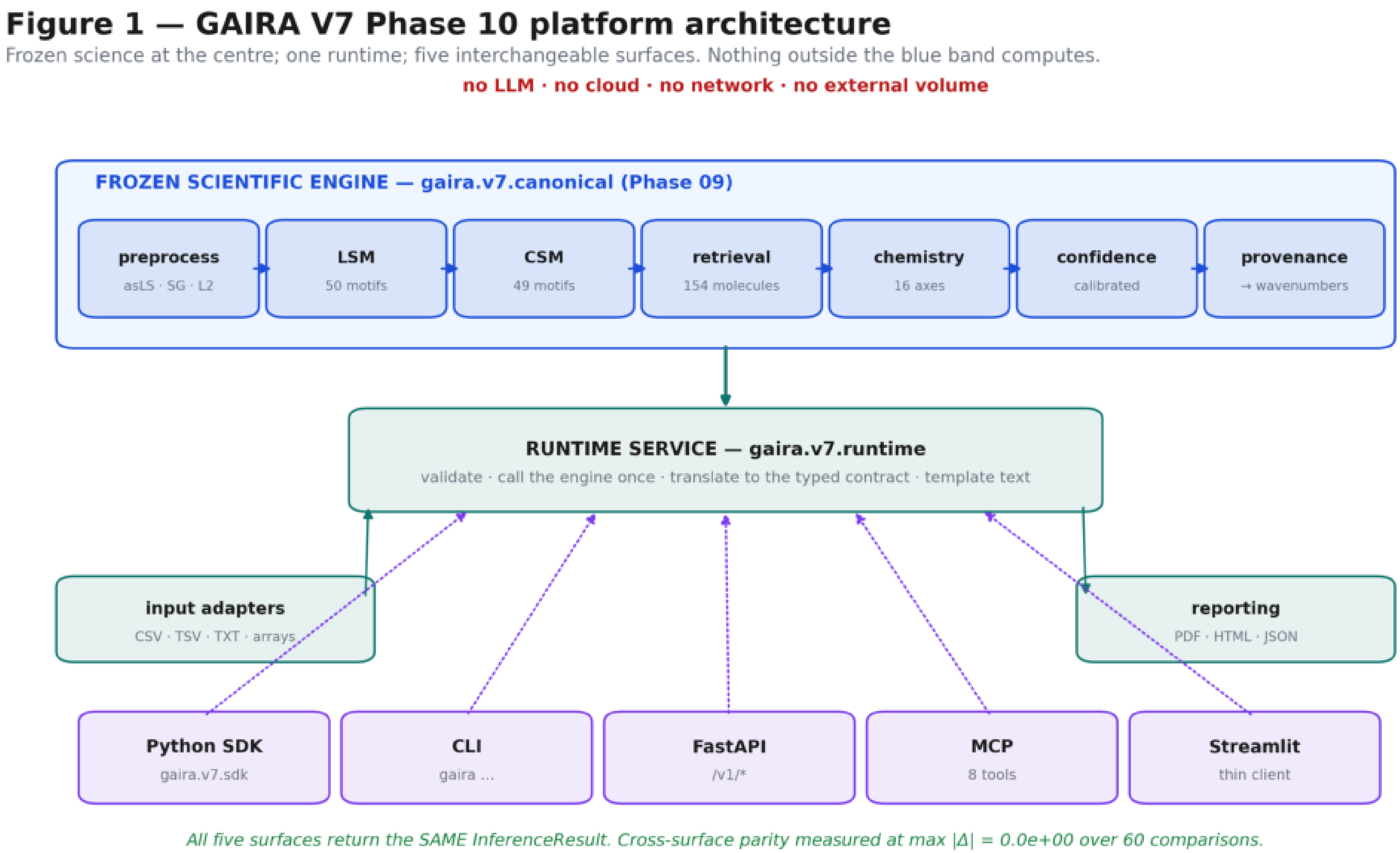
The green band is the runtime service. It validates input, calls the engine once, translates the engine's dictionaries into a typed public contract, and renders deterministic template text. It orchestrates and copies; it does not calculate.

The five purple boxes are the surfaces. Each is a different way of reaching the same runtime, and each returns the same object. A client may choose to display less; it may never compute more.

In plain terms

The line at the bottom is the phase's central measurement: across 60 comparisons spanning six independent code paths, the maximum absolute difference between any two surfaces was zero. Not 'within tolerance' — identical.

Figure 01



The complete Phase 10 platform. Frozen science at the centre, one runtime beneath it, five interchangeable surfaces below that. Nothing outside the blue band computes, and a static test fails the build if anything tries.

FIGURE 2

One rule, four layers

The rule is P-19: one implementation of every scientific quantity. If a number appears in two places it will eventually disagree in two places, and the disagreement will be discovered by a user rather than by a test.

The engine may compute. It is frozen, fingerprint-verified on every load, and nothing above it is permitted to reproduce any part of it.

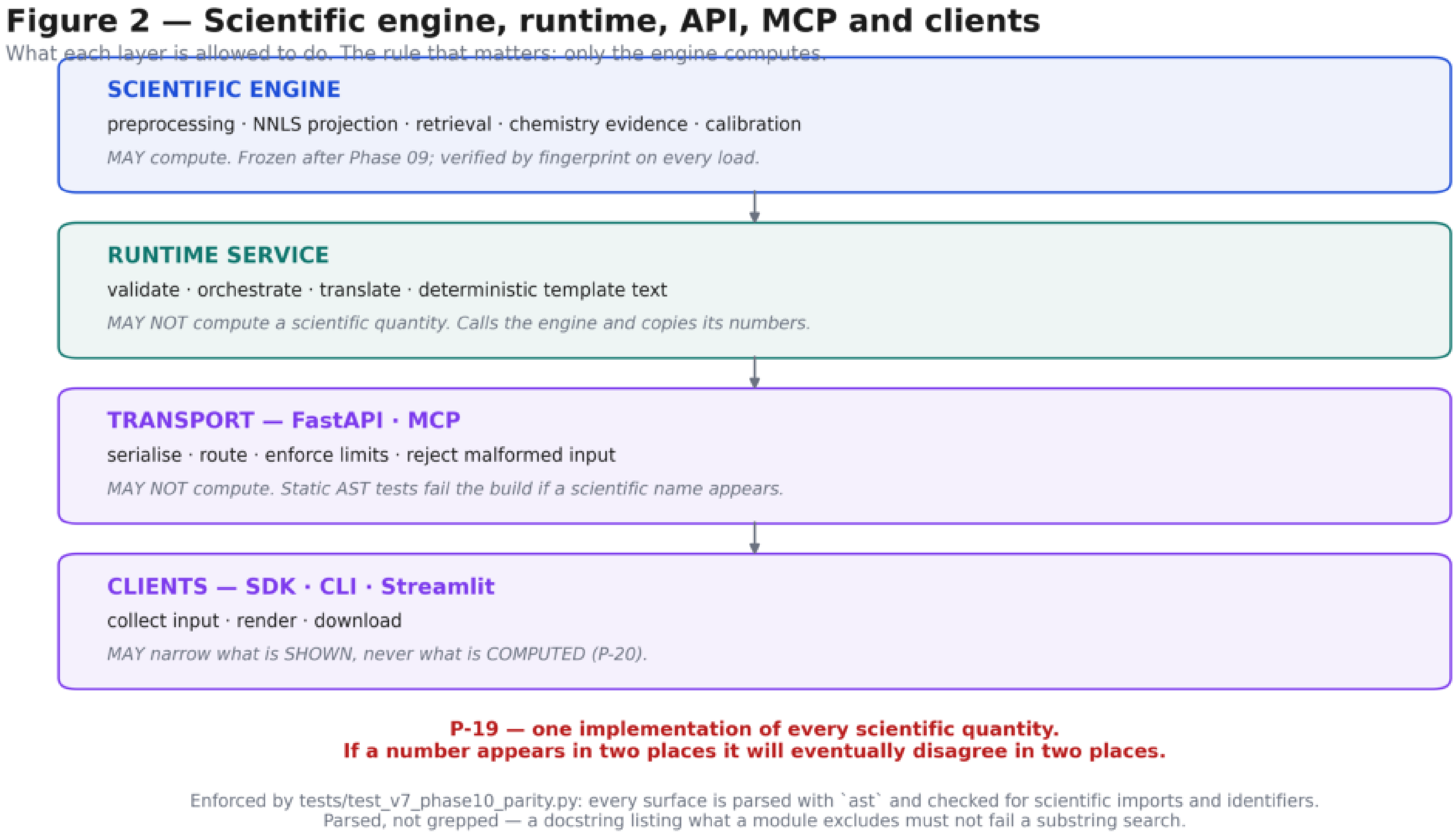
The runtime may validate, orchestrate and translate. It may not project, fit or calibrate. A static test parses it and asserts that it imports neither scipy nor scikit-learn and references no scientific primitive.

The transport and client layers may serialise, route, enforce limits, collect input and render. The Streamlit app additionally may not import the engine directly: it must go through the runtime rather than reach past it.

In plain terms

This is enforced by parsing each surface with Python's `ast` module rather than searching its text. A module docstring that lists what the module excludes would fail a naive substring search — Phase 09 learned that, and Phase 10 inherited both the lesson and the fix.

Figure 02



What each layer may do. The engine computes; the runtime orchestrates; the transport serialises; the clients render. P-19 in one picture.

FIGURE 3

From spectrum to answer

Eleven stages, always in this order, with no branches and no parameter a caller can tune at inference time.

The first two stages belong to Phase 10: parse the file and validate it. Validation answers one question — can the frozen engine say anything useful about this input, and if so with what caveats — and it answers at three severities. An error stops the run; a warning limits the interpretation; an info note records scope.

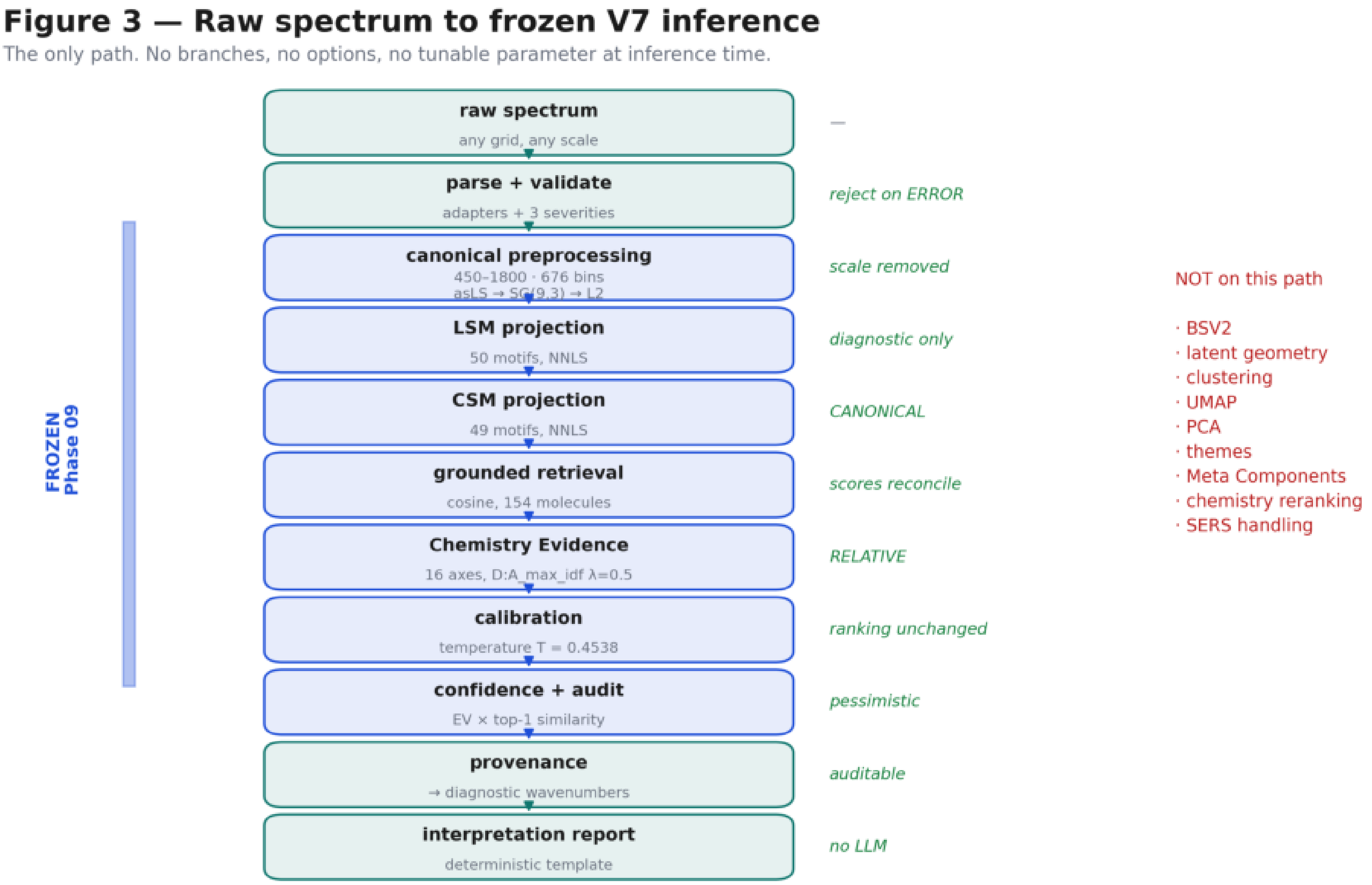
The seven blue stages are the frozen engine and were fixed by Phases 00 through 09. Canonical preprocessing removes absolute intensity, which is why nothing downstream can be read as a concentration. The CSM activation is the canonical representation — 49 numbers that every later stage reads and the only ones it reads.

The final stages produce confidence, provenance and a deterministic interpretation paragraph. No language model is involved at any point.

In plain terms

The red column is as important as the blue one. BSV2, latent geometry, clustering, UMAP, PCA, themes, Meta Components and chemistry-aware reranking are each absent because a measurement said so, not because of a preference. Four independent attempts to build a layer above the CSM each lost information.

Figure 03



The only inference path, stage by stage, with what is deliberately absent listed beside it. Every exclusion is a measured decision from an earlier phase.

FIGURE 4

The contract

Everything crossing the boundary between the frozen engine and the outside world is typed, versioned and free of NumPy. Arrays cross as lists of floats, so a JSON round-trip is lossless and no caller needs to know how the engine is implemented.

A request carries three things: the spectrum, its metadata, and the options. The metadata records modality and sample type. Modality is enforced — anything but Raman is rejected. Sample type is recorded and warned about but never applied, and a test proves it by running the same spectrum under four sample types and checking the result digest does not move.

A result carries eleven blocks and a digest. The digest is an MD5 over the scientific fields alone, excluding timestamps and free text, and it is the anchor the whole parity validation turns on: two surfaces agreeing means their digests are equal.

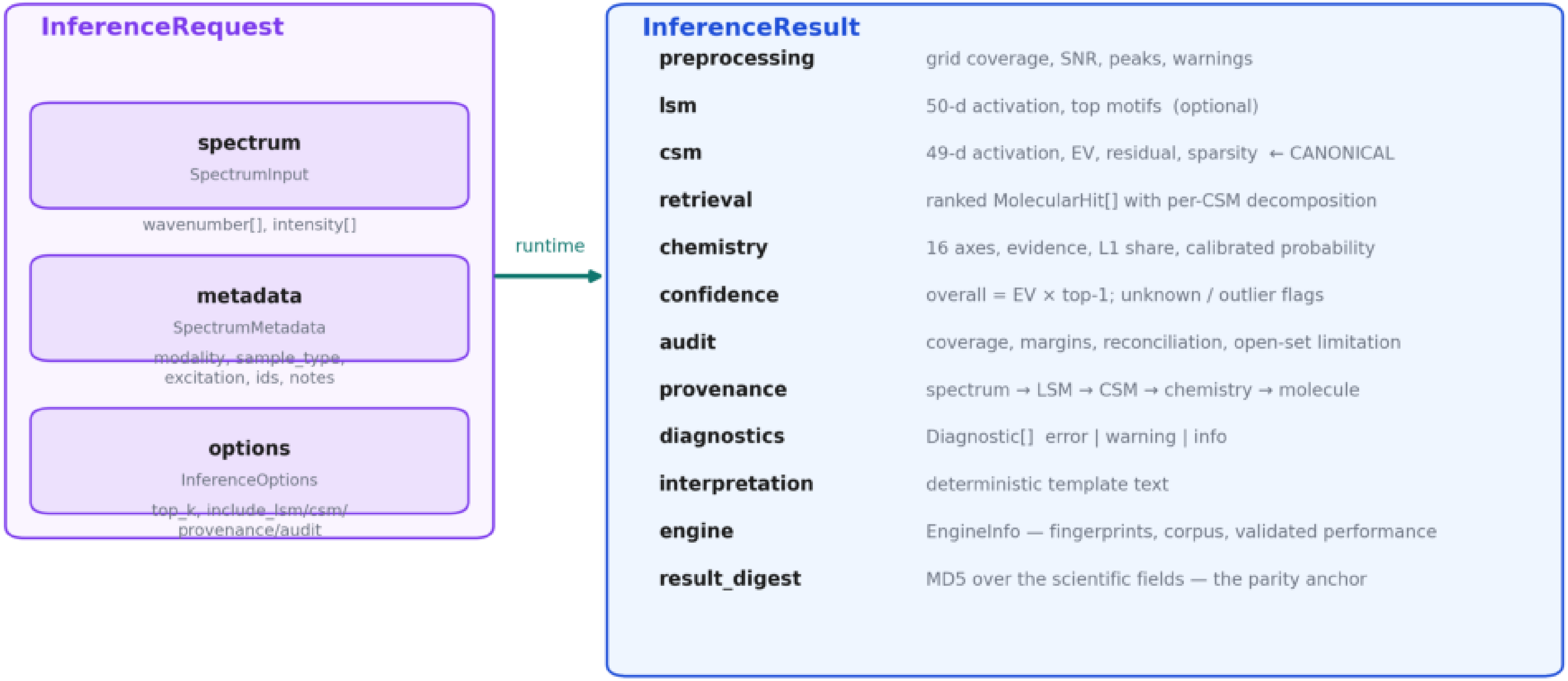
Every result object is a frozen dataclass. A record that can be edited after the fact is not a record.

In plain terms

One definition serves four consumers. The SDK returns these objects, the API serialises them, the MCP schemas are generated from the same shapes, and the report generator reads them. There is no second place where the response format is described, so there is no second place for it to drift.

Figure 4 — InferenceRequest and InferenceResult

The typed public contract. Pydantic v2 models serve the SDK, the API, the MCP schemas and the report generator from one definition.



No NumPy in the public contract — arrays cross as lists of floats and a JSON round-trip is lossless.
Every result is a frozen dataclass: a record that can be edited is not a record.

The typed public contract. One pydantic definition serves the SDK, the API, the MCP schemas and the report generator, so the four cannot drift apart.

FIGURE 5

HTTP

Six routes, versioned under /v1. Each does one thing and delegates to exactly one runtime method.

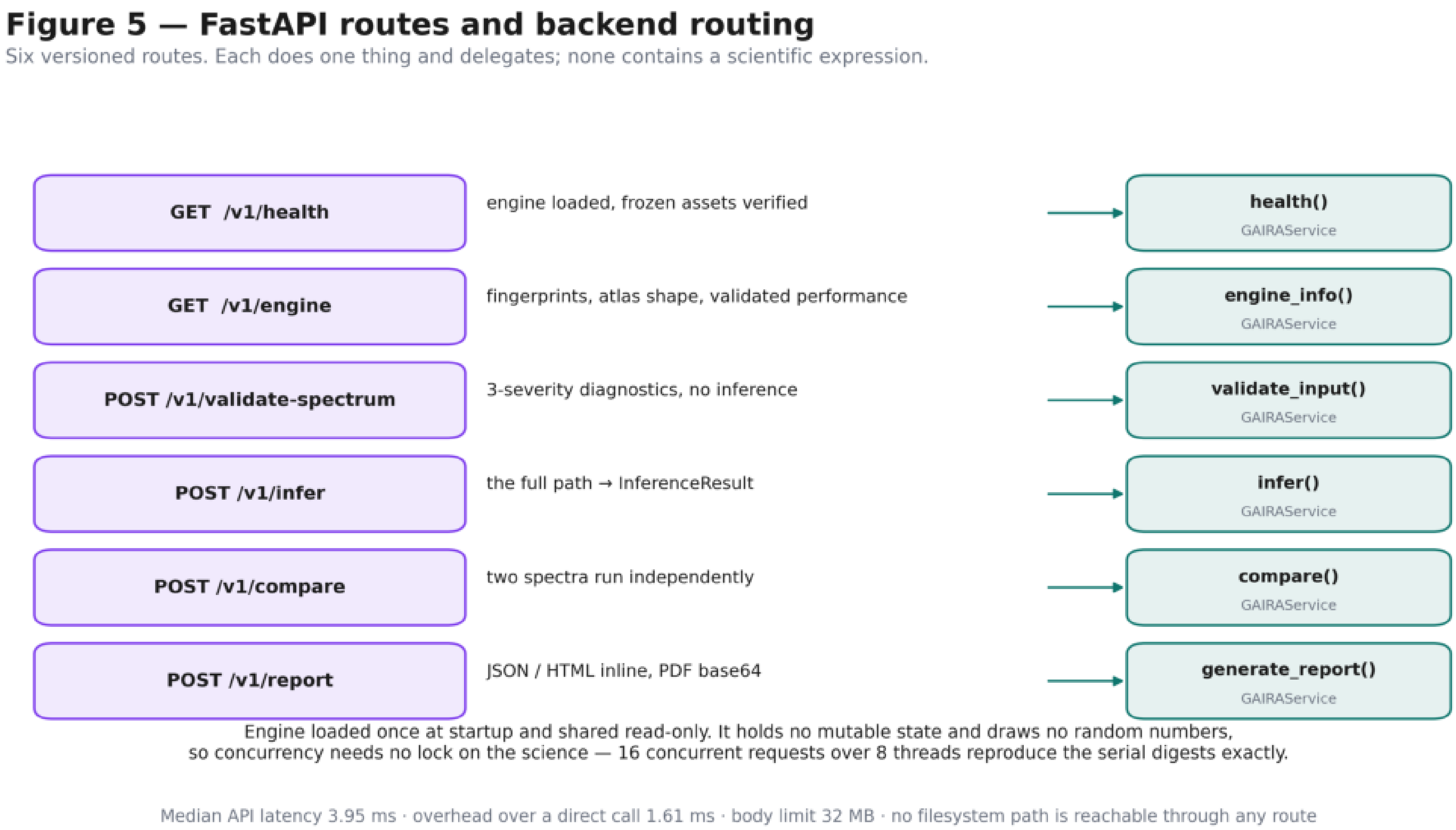
The engine loads once at startup, verifies ten frozen asset digests, and is shared read-only for the life of the process. Because it holds no mutable state and draws no random numbers, concurrency needs no lock on the science — and this was checked rather than assumed: sixteen requests across eight threads reproduce the serial digests exactly.

Validation failures return 422 with the complete diagnostic list, so a caller learns precisely which condition failed rather than receiving a generic rejection. Unknown request fields are rejected rather than ignored, which catches a typo in an option name instead of silently using the default.

The report route never writes to a caller-supplied path. JSON and HTML come back inline, a PDF comes back base64-encoded, and the filename is derived from the result digest. A report endpoint that writes where the caller says is a file-write primitive wearing a scientific label.

In plain terms

API overhead over a direct Python call is 1.6 ms against a 2.3 ms inference. The transport is not where the time goes.



Six versioned FastAPI routes, each delegating to one runtime method. No route contains a scientific expression and none accepts a filesystem path.

FIGURE 6

Tools, not primitives

The MCP server exposes the same runtime as eight callable tools. No language model runs inside it and it makes no network call; it is a provider, and whatever consumes it lives entirely outside the process.

The tools are deliberately coarse. An agent should be able to ask 'interpret this spectrum' and receive a scientifically coherent answer. It should not be able to request an NNLS solve against a raw dictionary matrix, because that is where a caller could construct a result the engine never sanctioned.

The narrow tools — chemistry evidence, molecular evidence, explain — return exactly the corresponding slice of the full inference for the same input. This is verified by digest equality rather than trusted, so a future edit that made one of them drift would fail a test.

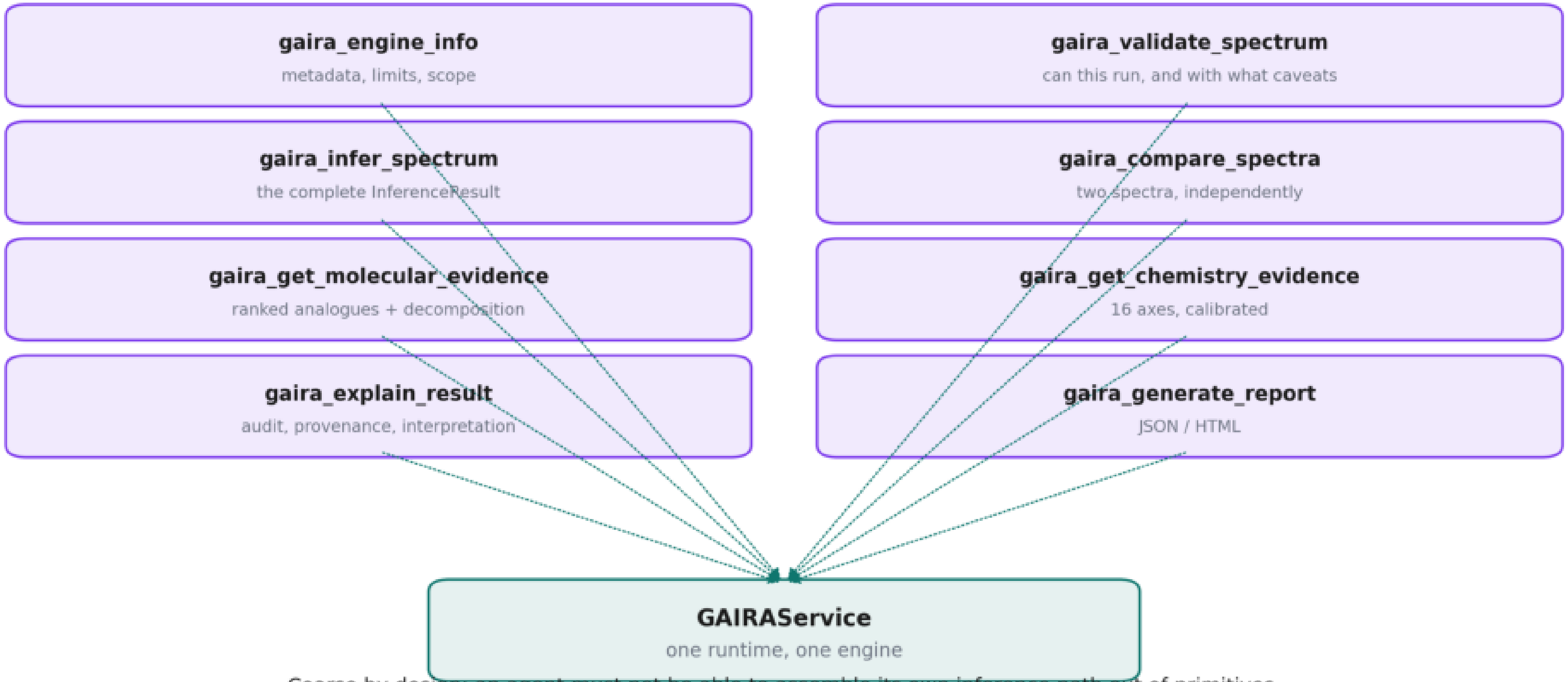
Every tool description carries its own caveats, and the server advertises the scope in its MCP instructions field, so a connecting client receives the limitations before its first call.

In plain terms

The engine_info tool should be called first and, in an agent deployment, required. It is the tool that states what the engine cannot do — no open-set detection, relative evidence only, reference analogues rather than identifications, Raman only.

Figure 6 — MCP tool orchestration

Eight read-only tools, deliberately coarse. The server provides tools; it runs no model.



Coarse by design: an agent must not be able to assemble its own inference path out of primitives.
No tool exposes NNLS, a raw matrix, or an arbitrary projection.

No LLM. No network. No cloud credential. The server is a provider; whatever consumes it lives entirely outside this process.

Eight MCP tools, deliberately coarse. An agent can ask for an interpretation; it cannot assemble its own inference path out of primitives.

FIGURE 7

The browser client

Seven hundred lines of Streamlit containing no scientific computation whatsoever. It uploads, calls the runtime, and renders.

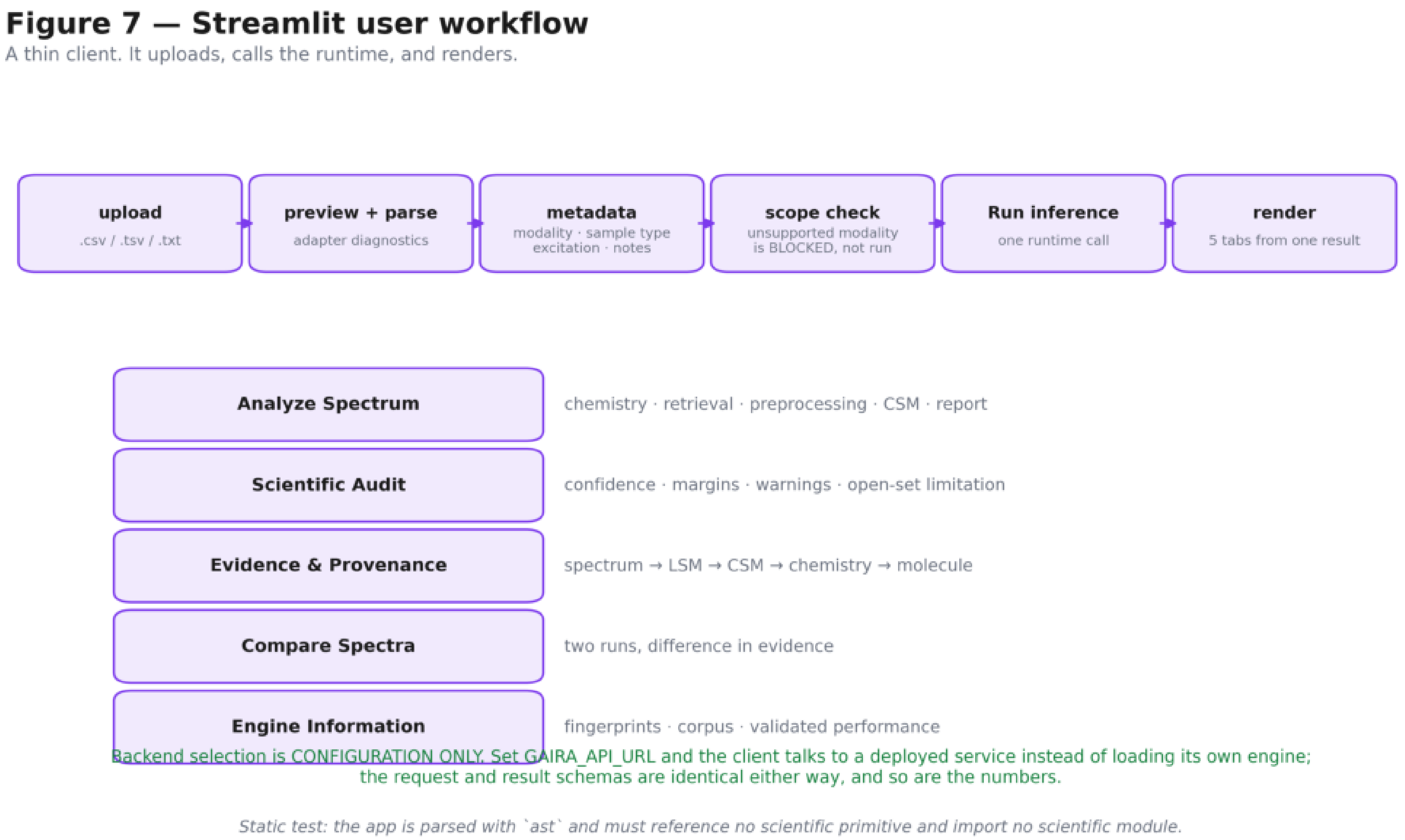
Backend selection is configuration only. With GAIRA_API_URL unset the app loads its own engine; with it set the app talks to a deployed service. The request and result schemas are identical either way and so are the numbers, so moving from a laptop to a server is an environment variable rather than a rewrite.

The processed spectrum and the CSM reconstruction that the app draws are returned by the engine, through a read-only accessor added in this phase. The UI never reproduces a transformation, so what a user sees is exactly what the projection consumed.

Language is constrained deliberately. The retrieval tab is titled Grounded Evidence Retrieval, not Molecule Identification. Every chemistry view carries the relative-evidence caveat. Selecting a non-Raman modality shows a red block and prevents the run rather than warning and proceeding.

In plain terms

A static test parses the app and fails the build if it references a scientific primitive, imports scipy or scikit-learn, or imports the engine directly. The UI must go through the runtime, not reach past it.



The Streamlit workflow and its five pages. Backend selection is configuration only — local engine or deployed API, identical numbers either way.

FIGURE 8

Why an answer is auditable

The provenance chain is what separates a GAIRA answer from a produced number.

Each retrieval score is a cosine similarity, which is an inner product of unit vectors. That means it decomposes exactly into per-motif terms — 'how much did each consensus motif contribute to this match'. The engine asserts that the parts sum to the whole within 1e-9 for every candidate it reports, on every call.

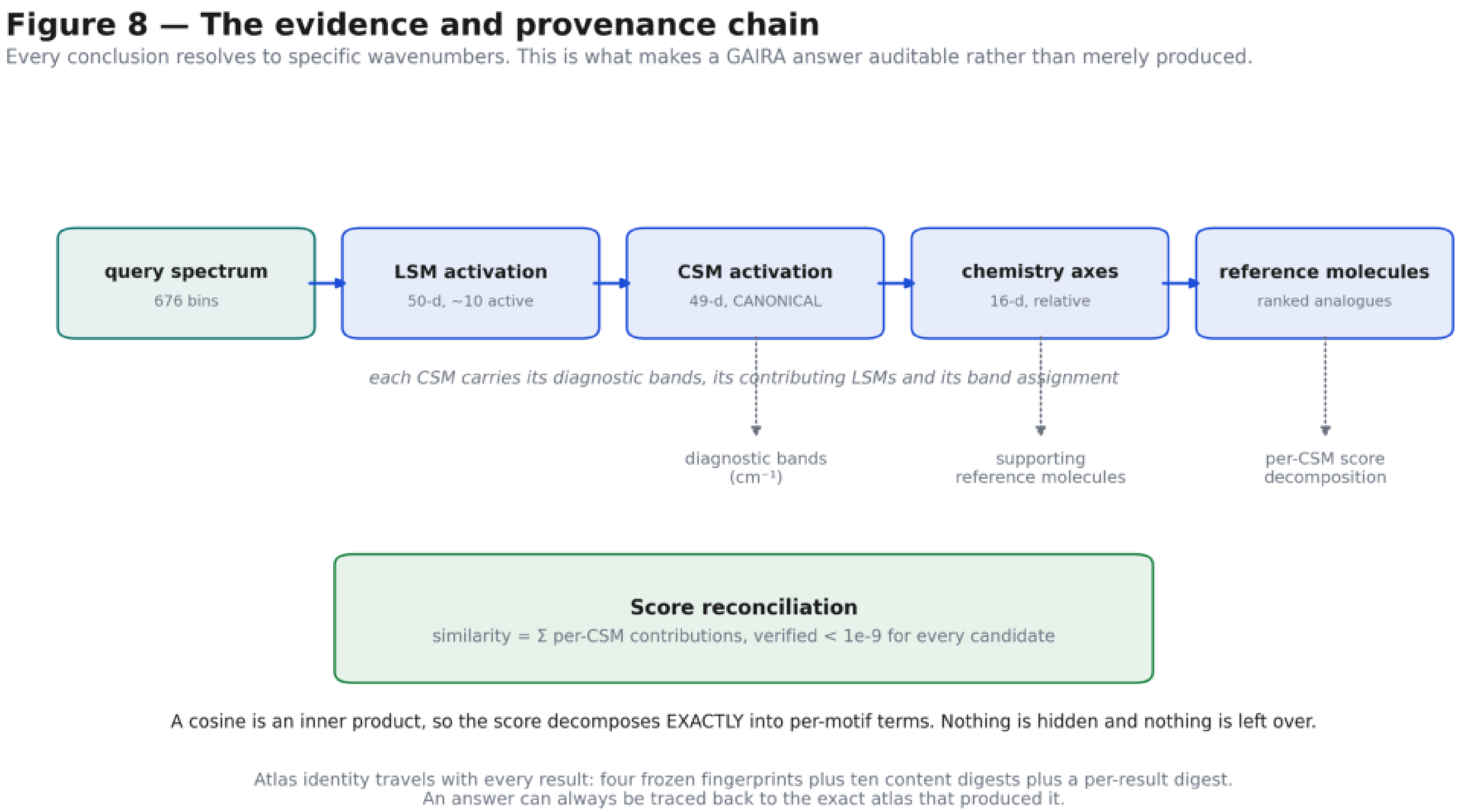
Each consensus motif carries its diagnostic band positions, its band assignment string, and the local motifs that formed it. So a chemistry conclusion can be walked back to specific wavenumbers, and a user can ask which part of the spectrum is responsible for a claim.

Atlas identity travels with every result: four frozen fingerprints, ten content digests, and a per-result digest. An answer can always be traced back to the exact atlas that produced it, and an atlas that has changed will not load.

In plain terms

Phase 10 added the second freeze layer. Phase 09 verifies four fingerprints recorded inside state files; Phase 10 pins the content of all ten files the engine opens. The first answers 'did the producing phase claim this artefact', the second answers 'is the file still the one that phase wrote'.

Figure 08



The provenance chain. A cosine is an inner product, so every retrieval score decomposes exactly into per-motif contributions, verified below 1e-9.

FIGURE 9

Extending to new measurement channels

Four modality adapters are declared and one is implemented. Every unimplemented adapter raises, with a statement of what a working version would have to supply.

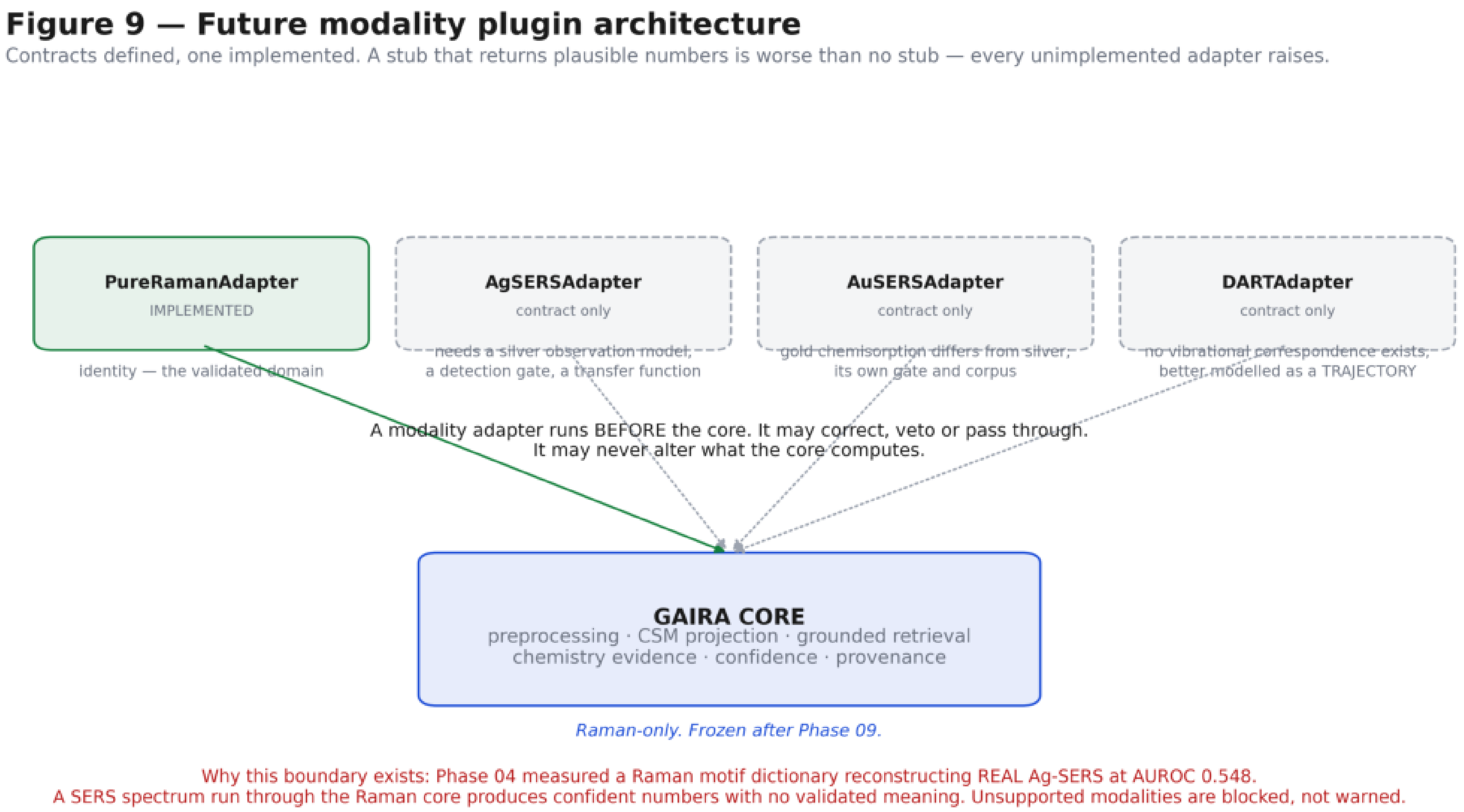
A modality adapter models the physics between the sample and the spectrum. It runs before the core and may correct, veto, or pass through — but it may never touch the dictionaries, the retrieval, or the chemistry model.

The reason this boundary is hard rather than advisory is a measurement. Phase 04 tested whether the frozen Raman atlas could detect real Ag-SERS as out of domain and got AUROC 0.548 — chance. A non-negative Raman motif basis reconstructs SERS of the same metabolites comfortably, so a SERS spectrum run through the Raman core produces confident numbers with no validated meaning.

So unsupported modalities are blocked rather than warned about. On the API that is a 422; in the SDK an exception; in the UI a red block that prevents the run.

In plain terms

DART is the interesting case. It has no vibrational correspondence to Raman, so it is not a spectral transform at all. It is better modelled as a trajectory over an orthogonal measurement — downstream of the representation rather than upstream of it — and the adapter says so instead of pretending otherwise.



Modality plugins: one implemented, four declared. Every stub raises, because a stub that returns plausible numbers is worse than no stub at all.

FIGURE 10

Extending to new sample contexts

Seven context adapters are declared and one is implemented. A context adapter frames a completed result; it runs after the core and cannot reach back into it.

The protocol is the enforcement. A context adapter returns caveats, framing and diagnostics — there is no field through which it could return evidence, so there is no way for it to rewrite a number. A test asserts the shape of that return type.

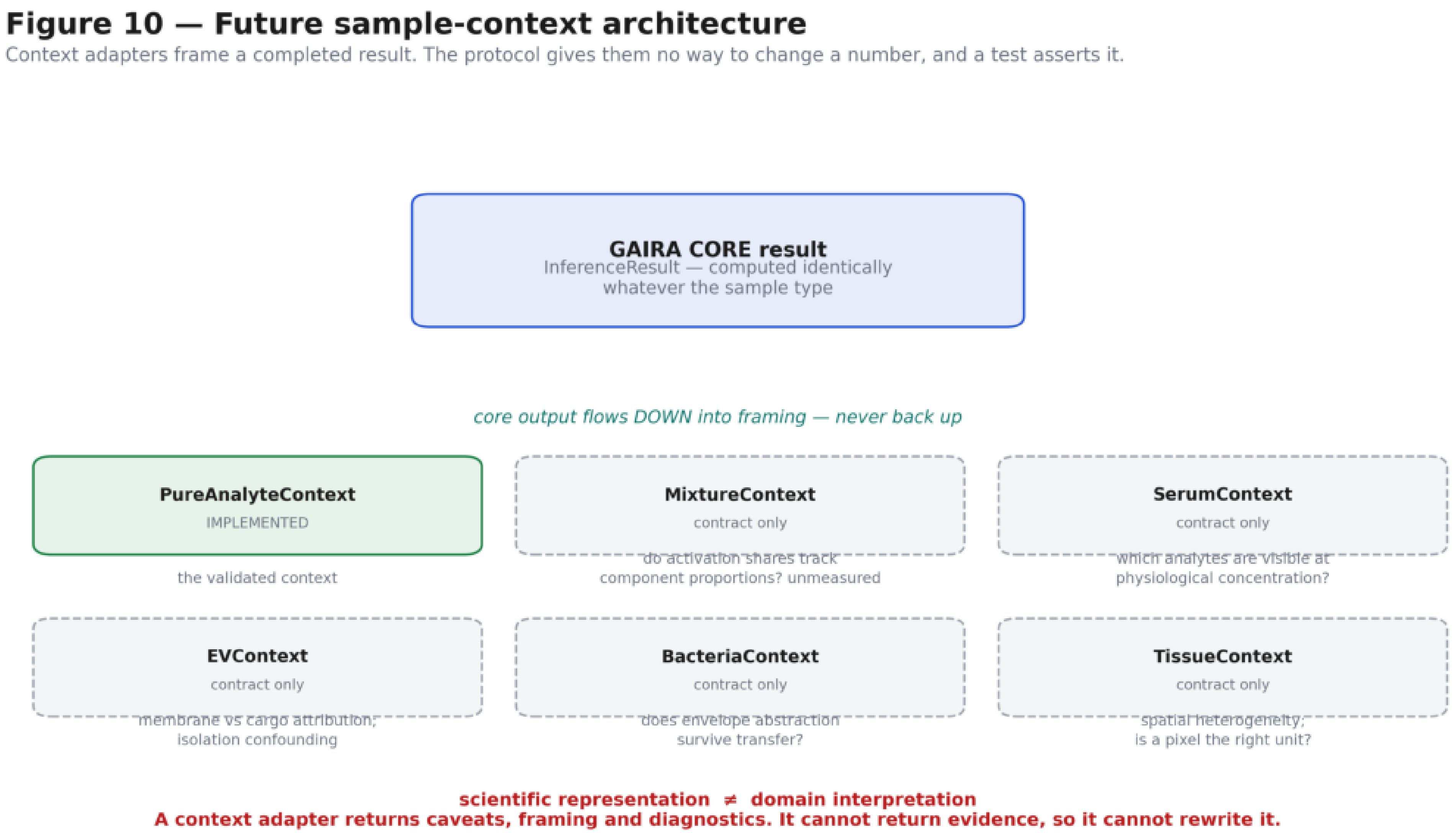
Each unimplemented context states the open questions a working version must answer. For serum: which analytes are visible at physiological concentration, and how albumin dominance should be handled. For extracellular vesicles: membrane versus cargo attribution, and isolation-method confounding. For tissue: whether a pixel is even the right unit.

Meanwhile sample_type is accepted today as metadata. It produces a scope warning and changes nothing in the calculation, and a test runs the same spectrum under four sample types to prove the digest does not move.

In plain terms

The principle this figure encodes: scientific representation is not domain interpretation. Confusing the two is how a spectral engine becomes a clinical claim without anyone deciding that it should.

Figure 10



Sample-context plugins. Context adapters frame a completed result; the protocol gives them no way to change a number.

FIGURE 11

Where an agent would sit

Phase 10 ships no language model and requires no cloud account. This figure is the boundary a Phase 11 agent layer would have to respect.

An agent sits above the tool server. It may choose which tools to call, ask for more evidence, rephrase the deterministic interpretation, compare results it obtained through the tools, and surface the caveats verbatim.

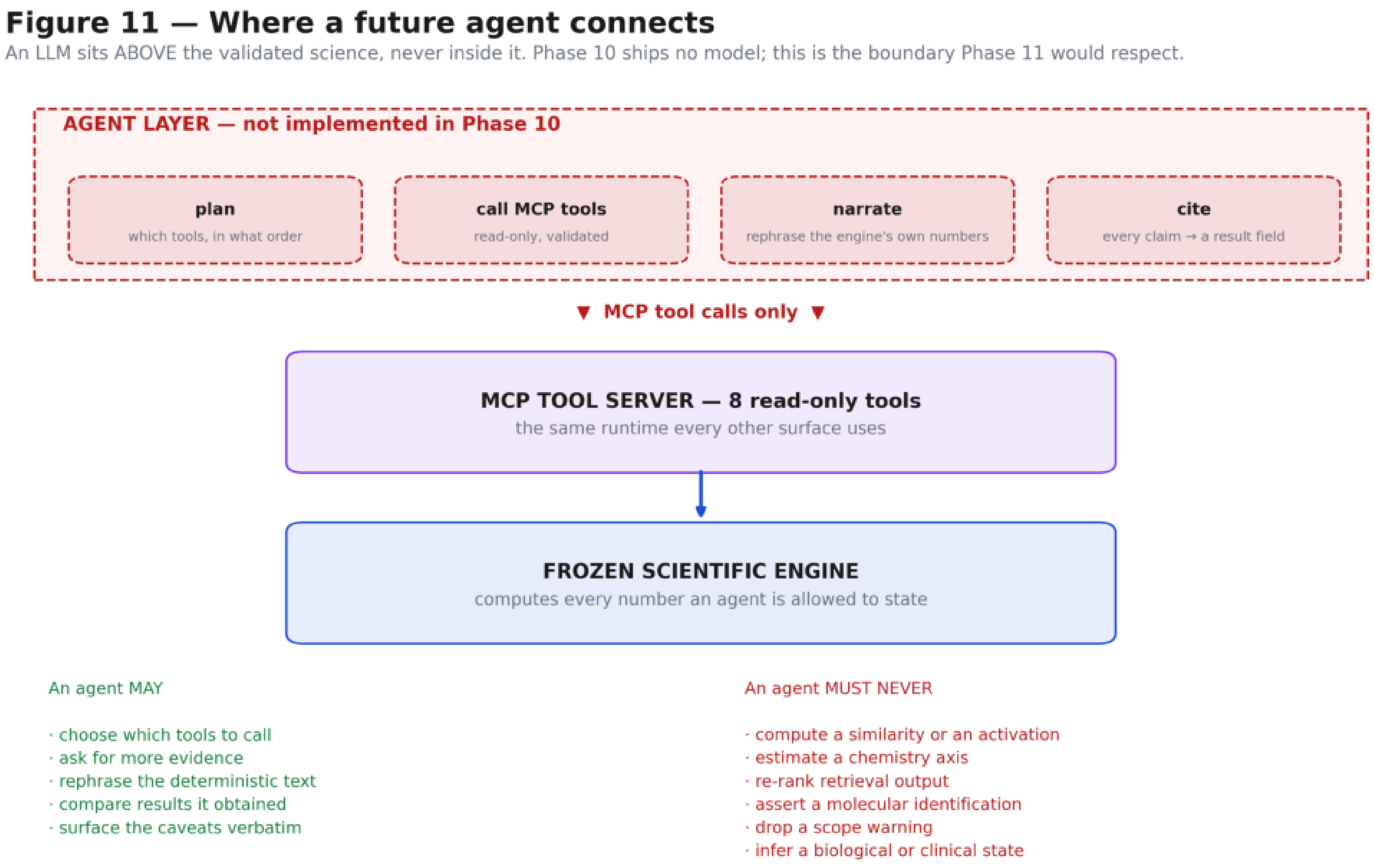
It may never compute a similarity, an activation, a chemistry axis or a confidence; re-rank retrieval output; assert a molecular identification; convert relative evidence into a concentration; treat low confidence as evidence of novelty; drop a scope warning; or infer a biological or clinical state.

Every number an agent states must be traceable to a field of an InferenceResult, and every claim about a molecule must carry the word analogue or an equivalent.

In plain terms

The honest assessment is that the engineering guardrails are in place and the linguistic one is not. Every failure mode this architecture prevents is a failure of language, and an agent is a language system. Phase 11 should open with an adversarial overclaim benchmark, not close with one.

Figure 11



Where a future agent connects — above the tool server, never inside the science. Phase 10 ships no model; this is the boundary Phase 11 would respect.

FIGURE 12

The measurement the phase turns on

Twelve locked spectra through six independent surfaces, compared field by field at a tolerance of one part in a trillion.

The spectra were chosen for behaviour, not convenience: the highest-confidence correct case, a chemistry-right but molecule-wrong case, the lowest-explained-variance spectrum in the entire corpus, an ambiguous class, four different chemistry families, three different source libraries, and a synthetic noise control.

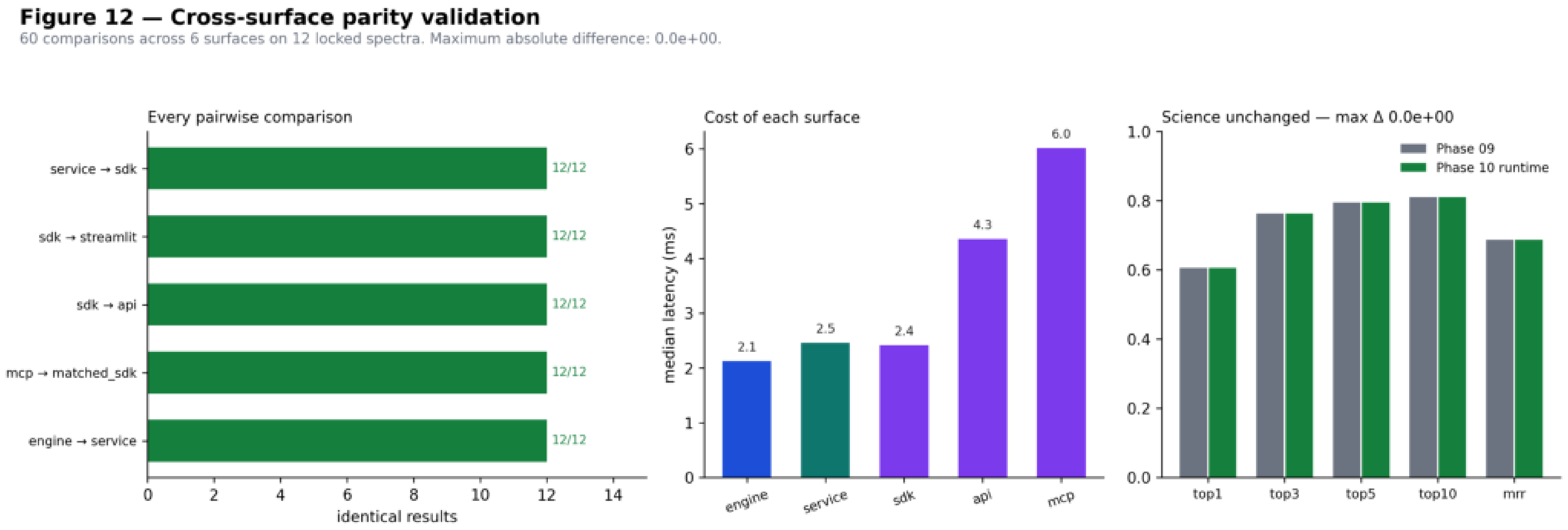
Compared per case: all 49 CSM activations, explained variance, all 16 chemistry evidence values, all 16 calibrated probabilities, the predicted class, all ten retrieval ranks and similarities, overall confidence, the unknown flag, grid coverage, and score reconciliation.

Sixty comparisons, zero divergent, maximum absolute difference exactly zero. The right panel shows the other half of the obligation: every Phase 09 retrieval figure reproduced through the runtime path at deviation 0.0.

In plain terms

The middle panel is worth reading as a cost sheet. Reaching the engine through HTTP costs 1.6 ms and through MCP 3.0 ms, against a 2.3 ms inference. The abstraction is not where the time goes, so there is no performance argument for bypassing it.

Figure 12



Cross-surface parity: 60 comparisons across six surfaces, zero divergent, maximum absolute difference 0.0, and every Phase 09 retrieval figure reproduced exactly.

Validation

Phase 10 has one scientific obligation: prove that packaging did not change the science. It is met to the digit.

PART III

The engine freeze audit

Nine gates. Every fingerprint recomputed from the committed tree rather than read from documentation.

gate	status
FA1 every frozen asset present and content-pinned	PASS
FA2 declared fingerprints recomputed and match	PASS
FA3 canonical ontology ordering unchanged	PASS
FA4 engine deterministic on repeat	PASS
FA5 golden fixtures stored for six representative cases	PASS
FA6 Phase 09 retrieval reproduced within 1e-6	PASS
FA7 Phase 09 chemistry reproduced within 1e-6	PASS
FA8 no frozen artefact written or modified	PASS
FA9 engine reads only committed repository assets	PASS

Phase 09 reproduction, all 375 spectra

metric	Phase 10	Phase 09	deviation
top1	0.605333	0.605333	0.0e+00
top3	0.762667	0.762667	0.0e+00
top5	0.794667	0.794667	0.0e+00
top10	0.810667	0.810667	0.0e+00
mrr	0.687003	0.687003	0.0e+00

Maximum deviation 0.0e+00. Not 'within tolerance' — identical.

PART III

Cross-surface parity

60 comparisons across 6 surfaces on twelve locked spectra, at a tolerance of 1e-12.

gate	status
P1 six surfaces produce identical scientific output	PASS
P2 result digests agree across service, SDK, API and Streamlit paths	PASS
P3 malformed input rejected on every surface	PASS
P4 unsupported modality blocked on every surface	PASS
P5 concurrent API requests reproduce serial results	PASS
P6 Phase 09 science reproduced through the runtime path	PASS
P7 single-spectrum inference is interactive (< 250 ms)	PASS
P8 API overhead under 100 ms	PASS

Scientific validation through the runtime path

metric	Phase 10 runtime	Phase 09	deviation
molecule top1	0.605333	0.605333	0.0e+00
molecule top3	0.762667	0.762667	0.0e+00
molecule top5	0.794667	0.794667	0.0e+00
molecule top10	0.810667	0.810667	0.0e+00
molecule mrr	0.687003	0.687003	0.0e+00
chemistry top1 in sample	0.954667	0.954667	0.0e+00
csm mean explained variance	0.823196	0.823196	0.0e+00

In plain terms

Six independent code paths were given the same spectra and compared field by field: 49 activations, 16 evidence values, 16 calibrated probabilities, ten ranks and similarities, confidence, coverage and reconciliation. The largest disagreement anywhere was zero.

PART III

Performance

Measured on a laptop before any gate was written. No latency threshold was invented in advance of the numbers.

engine load seconds	0.279
single inference ms median	2.34
single inference ms p95	3.11
ten sequential seconds	0.021
hundred sequential seconds	0.218
api ms median	3.95
api overhead ms	1.61
mcp ms median	5.3
mcp overhead ms	2.97
report pdf seconds	1.417
report json seconds	0.002
max rss mb	595.5

At 2.3 ms per spectrum the engine sustains roughly 430 spectra per second single-threaded. For live Raman use, acquisition is the bottleneck, not inference.

In plain terms

The cost of reaching the engine through HTTP is 1.6 ms and through MCP 3.0 ms, against a 2.3 ms inference. There is no performance argument for bypassing the runtime, which matters: the usual reason a client ends up reimplementing science is that the abstraction felt expensive.

Defects found and fixed

Four defects were found and fixed during this phase. Two of them are the same defect.

1 · The freeze audit reimplemented the science

The first script Phase 10 wrote hand-rolled the leave-one-out retrieval loop and dropped every spectrum of the query molecule instead of only the query spectrum.

Had it stood: It reported molecule top-1 of 0.0000 — an apparent catastrophic regression that was entirely an artefact of the audit. Chasing it would have wasted the phase; believing it would have falsely condemned the engine.

Fix: Replaced with calls to the frozen retrieval modules. Deviation dropped to exactly 0.0.

2 · The text adapter desynchronised its columns

A row whose wavenumber parsed and whose intensity did not appended the wavenumber and then failed, pairing every subsequent intensity with the wrong wavenumber.

Had it stood: A silently mangled spectrum that still looks like a spectrum — the worst failure an adapter has, because nothing downstream can detect it.

Fix: Parse both values before appending either, with a regression test that checks the pairing arithmetic explicitly.

3 · A broad exception handler hid defect 2

The adapter loader swallowed parse exceptions and reported 'unrecognised format' for a file it had actually accepted.

Had it stood: Defect 2 was invisible for as long as this stood, and it presented as a format problem, so the investigation would have started in the wrong place.

Fix: Only the cheap recognition step is guarded now; a failing parse returns an explicit diagnostic naming the adapter and the exception.

4 · The static test flagged its own documentation

A substring search for a banned term matched the Streamlit docstring listing what the file excludes.

Had it stood: A permanently red test that would be disabled rather than fixed — and then the rule it encodes would be gone.

Fix: AST-based checking of referenced names and imported modules, ignoring strings and comments.

Note

Defect 1 matters out of proportion to its size. The very first script this phase wrote reproduced the exact failure mode the phase exists to prevent — reimplementing scientific logic outside the engine — within twenty minutes of starting. That is why the one-implementation rule is enforced by tests rather than stated as a principle: the discipline does not survive contact with convenience.

IV

Extension and the agent boundary

How new measurement channels and sample contexts would attach, and where a language model may and may not sit.

PART IV

The plugin contracts

Four extension protocols are defined. One modality adapter and one context adapter are implemented; the other ten raise.

protocol	runs	may do	may never do
ModalityAdapter	before the core	correct, veto, pass through	touch the dictionaries
SampleContextAdapter	after the core	add caveats and framing	change a number
InterpretationAdapter	after the core	rephrase	compute anything
TrajectoryAdapter	over a sequence	analyse change	recompute a result

Why no stub returns a number

A stub that produces plausible output is worse than no stub at all: it will be wired up, its results will be plotted, and nobody will remember it was never validated. Every unimplemented adapter raises with a statement of what a working version must supply — an observation model, a detection gate, a transfer function, and its own held-out corpus.

Why the modality boundary is hard rather than advisory

Phase 04 tested whether the frozen Raman atlas could detect real Ag-SERS as out of domain. It got AUROC 0.548 — chance. A non-negative Raman motif basis reconstructs SERS of the same metabolites comfortably, so a SERS spectrum run through the Raman core produces confident numbers with no validated meaning.

In plain terms

scientific representation is not domain interpretation. Confusing the two is how a spectral engine becomes a clinical claim without anyone deciding that it should.

Where a future agent connects

Phase 10 ships no language model and requires no cloud account. This is the boundary a Phase 11 agent layer would have to respect.

An agent MAY

- choose which tools to call, and in what order
- ask for more evidence before answering
- rephrase the deterministic interpretation
- compare results it obtained through the tools
- surface the caveats verbatim

An agent MUST NEVER

- compute a similarity, an activation, a chemistry axis or a confidence
- re-rank retrieval output
- assert a molecular identification
- convert relative evidence into a concentration or an abundance
- treat low confidence as evidence of a novel molecule
- drop a scope warning
- infer a biological or clinical state

What must be validated first

- an adversarial overclaim benchmark — real tool output plus a leading question, measuring how often the model overclaims
- a citation check: every numeric claim resolves to a result field
- a caveat-retention check: scope, relative-evidence and analogue caveats survive into the output
- a refusal check: the agent declines questions the engine cannot support
- a determinism boundary: the report generator remains the system of record, agent text is marked as commentary

Note

The engineering guardrails are in place; the linguistic one is not. Every failure mode this architecture prevents is a failure of language — calling an analogue an identification, calling relative evidence a concentration, calling low confidence a discovery. An agent is a language system, and those are exactly the sentences it would find natural to produce.

V

Scope, limits and the decision

What this platform may and may not be used to claim.

Interpretation and engineering limits

Constraints on what the output MEANS, not caveats about its quality.

— Chemistry Evidence is RELATIVE — not a concentration, not an abundance, not a mixture fraction. L2 normalisation removes absolute scale in the first preprocessing stage.

— Retrieved molecules are reference ANALOGUES, not identifications. Validated molecule top-1 is 0.6053, and 68 of 375 corpus queries are unretrievable by construction.

— There is NO validated open-set detection. The engine cannot determine that the true molecule is absent from its bank; white noise reconstructs at CSM explained variance 0.61, above the 0.50 warning floor. Read the confidence, not the flag.

— Pure Raman reference spectra only. SERS, serum, plasma, EV, bacteria and tissue behaviour is unmeasured. Contracts exist; validation does not.

— The 16 chemistry classes are a curated cut through a continuum, not a discovered structure.

— Class-prior bias remains open — class sizes range from 3 to 80 spectra.

— In-sample chemistry figures describe the shipped fit. Quote the held-out 0.8507.

Engineering limits

— No real instrument export has been through the adapters — Renishaw, B&W Tek and Horiba files were unavailable.

— Validation thresholds are reasoned and declared in advance, but not swept; no false-positive rate is attached.

— No production hardening: no authentication, rate limiting, TLS or audit log. Loopback by default.

— PDF bytes depend on the matplotlib version; content is reproducible, bytes are reproducible on a fixed environment.

PART V

The decision

All seventeen gates pass. No defect remains open.

Frozen Phase 09 engine verified	PASS — 9/9, fingerprints recomputed
Scientific outputs unchanged	PASS — max deviation 0.0e+00
Cross-surface numerical parity	PASS — max discrepancy 0.0e+00
No duplicated scientific logic	PASS — AST-enforced on five surfaces
No LLM or cloud dependency	PASS — statically verified
Raman-only scope preserved	PASS — blocked on all five surfaces
Extension contracts defined	PASS — 11 declared, 10 raise
Local clean-clone inference	PASS — 10 committed files
SSD_Rad required?	NO
Full test suite	1436 passed · 1 skipped · 0 failed

Recommendation

FREEZE THE GAIRA V7 RUNTIME. Ready to design a Phase 11 agent layer, conditional on the five linguistic validations in Part IV.

Assessment

Confidence that packaging changed no science: 10/10 — checkable, and checked at deviation 0.0. Confidence that the platform is stable enough to build on: 9/10. The deduction is the adapters: no real instrument export has been parsed, and file handling is the one part of this platform whose correctness cannot be argued from first principles.

Colophon and provenance

Every figure, table and number in this document was generated from the committed Phase 10 artifacts by the scripts listed below. No value was typed by hand.

engine	src/gaira/v7/canonical/engine.py
runtime	src/gaira/v7/runtime/
freeze audit	results/v7_rebuild/phase10/code/run_engine_freeze_audit.py
parity + performance	results/v7_rebuild/phase10/code/run_parity_and_performance.py
figures	results/v7_rebuild/phase10/code/make_figures.py
this document	results/v7_rebuild/phase10/code/make_pdf.py
tests	tests/test_v7_phase10_{runtime,surfaces,parity}.py

atlas fingerprint	2e43ddcca7d3be41c5f9da016fb8277f
frozen atlas	09ed804a40836f4a05a91ba10900cded
frozen LSM registry	208482d6f7178b5b8f16cace91be55b0
frozen CSM registry	0b4aa550ccefed3edabdbde5bae11c8d
frozen Phase 05 engine	20d8bd99ce71f45a125c6a2b1d719e51

Companion documents

- PHASE_10_CONTEXT.md — the continuity note
- PHASE_10_ENGINE_FREEZE_AUDIT.md — fingerprints, golden fixtures, regression
- PHASE_10_ARCHITECTURE.md — packages, rules, freeze layers
- PHASE_10_API_SPEC.md · PHASE_10_MCP_SPEC.md · PHASE_10_STREAMLIT_SPEC.md
- PHASE_10_PLUGIN_ARCHITECTURE.md — the four extension contracts
- PHASE_10_DEPLOYMENT_GUIDE.md — clean clone to running in five minutes
- PHASE_10_VALIDATION_REPORT.md — all seventeen gates
- PHASE_10_SCIENTIFIC_AUDIT.md — strongly / weakly / unsupported
- PHASE_10_DECISION_GATE.md — the verdict